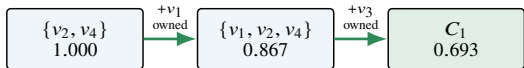


Feasible parent. For feasible D , delete the vertex $d(D)$ with minimum total similarity. Then $\text{avgSim}(D \setminus \{d(D)\}) \geq \text{avgSim}(D)$.

Unique ownership. Traverse $C \rightarrow D = C \cup \{x\}$ only if $x = d(D)$. Thus each state has one parent and needs no visited table.

Owned path to C_1



Two paths to C_2

Traverse

$\{v_5, v_6\} + v_3 \rightarrow C_2$
 $v_3 = d(C_2)$

Stop duplicate path

$\{v_3, v_5\} + v_6 \rightarrow C_2$
 $v_6 \neq d(C_2) = v_3$

Reverse search reaches each feasible state once;
the final containment filter returns C_1 and C_2 .